

Jonathan A Diller

Jonathan.A.Diller@Gmail.com | jond07.github.io | Updated: May 2025

Research Interests

My research interests broadly lie in multi-robot planning and tasking. More specifically, I focus on developing planning and tasking algorithms that account for how a robot's actions affect the state of its own systems, how the environment influences its behavior, and how the actions of individual robots impact their ability to coordinate within a team. A common theme in my work is the integration of realistic systems models into both centralized and decentralized algorithms, with an emphasis on bridging theory and practice. I also have a broader interest in Operations Research and Machine Learning.

Education

Colorado School of Mines <i>Doctor of Philosophy in Robotics (In Progress)</i>	Golden, CO <i>Expected Graduation: August 2025</i>
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Colorado School of Mines <i>Master of Science in Computer Science</i> GPA: 4.0/4.0	Golden, CO May 2022
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Pennsylvania State University, Harrisburg <i>Bachelor of Science in Computer Science</i> - Minor in Mechatronics Technology - Minor in Mathematics - Graduated summa cum laude - GPA: 3.99/4.0	Middletown, PA May 2020
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Research Experience

Research Assistant (<i>Under Dr. Qi Han</i>) <i>Pervasive Computing Systems Group, Colorado School of Mines</i> - Studying and implementing multi-robot projects with focus on communication and energy-aware planning. - Writing proposals (contributed to two white papers)	July 2020 - Present Golden, CO
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Journeyman Fellow (<i>Mentor: Dr. J. Humann and J. Dotterweich</i>) <i>Army Research Laboratory</i> Researched and developed planning algorithms for mixed drone-UGV teams.	June 2024 - August 2024 Golden, CO (remote)
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Graduate Research Intern (<i>Mentor: Dr. John Rogers</i>) <i>Army Research Laboratory</i> Researched the topics of task allocation and communication mapping for robot teams.	May 2023 - August 2023 Adelphi, MD
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Research Scholar (<i>Mentor: Dr. Peter Idowu</i>) <i>Penn State Drawdown REU Program</i> Designed and evaluated algorithms for controlling microgrids using PLCs.	June 2019 - July 2019 Middletown, PA
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Research Assistant (<i>Under Dr. Javad Khazaei</i>)	March 2019 - June 2019
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- Researched and developed microcontroller applications for use in renewable energy projects.

Publications

Under Review

- [R1] **J. Diller**, Q. Han, and M. Desaulniers[†]. “*Holistic Framework for Multi-Drone Data Collection in IoT Systems.*” Under review at the IEEE Internet of Things Journal.

Journal Papers

- [J1] **J. Diller**, J. Rogers, Q. Han, and N. T. Dantam. “*Communication Mapping for Robot Teams.*” Accepted for publication at the IEEE Transactions on Field Robotics, 2025.
- [J2] **J. Diller**, Q. Han. “*Energy-Aware Drone Path Finding with a Fixed-Trajectory Ground Vehicle.*” Published in the ACM Journal on Autonomous Transportation Systems, 2025.

Conference Papers

- [C1] **J. Diller**, Y. S. Teoh[†], R. Byers[†], Q. Han, J. G. Rogers, and N. T. Dantam. “*Failure-Aware Tasking for Teams of Drones.*” In proceedings of the International Conference on Distributed Computing Systems (ICDCS), 2025. *Acceptance rate: 19.7%.*
- [C2] **J. Diller**, Q. Han, R. Byers[†], J. Dotterweich, and J. Humann. “*Hitchhiker’s Guide to Patrolling: Path-Finding for Energy-Sharing Drone-UGV Teams.*” In proceedings of the International Conference on Autonomous Agents and Multiagent Systems (AAMAS), 2025. *Acceptance rate: 24.5%.*
- [C3] C. Wang, **J. Diller**, and Q. Han. “*LLM for Generating Simulation Inputs to Evaluate Path Planning Algorithms.*” In proceedings of the International Conference on Machine Learning and Applications (ICMLA), 2024. *Acceptance rate: 24.3%.*
- [C4] **J. Diller**, P. Hall[†], and Q. Han. “*Holistic Path Planning for Multi-Drone Data Collection.*” In proceedings of the International Conference on Distributed Computing in Smart Systems and the Internet of Things (DCOSS-IoT), 2023. *Acceptance rate: 38.4% (short paper).*
- [C5] **J. Diller** and Q. Han. “*Energy-Aware UAV Path Planning with Adaptive Speed.*” In proceedings of the International Conference on Autonomous Agents and Multiagent Systems (AAMAS), 2023. *Acceptance rate: 23.3%.*
- [C6] **J. Diller**, P. Idowu, J. Khazaei, “*Load-Leveling Trainer for Demand Side Management on a 45kW Cyber-Physical Microgrid,*” *Texas Power and Energy Conference 2020 (TPEC)*, 2020

- [C7] **J. Diller**, B. Trussell, J. Khazaei, P. Idowu, “*Hardware Development of a Three-Phase 3.5 kW SiC Converter with Sinusoidal PWM*,” *Texas Power and Energy Conference 2020 (TPEC)*, 2020

Workshop Papers

- [W1] **J. Diller**, N. Dantam, J. Rogers, and Q. Han. “*Communication Jamming-Aware Robot Path Adaptation*.” In proceedings of the International Workshop on Distributed Collective Intelligence (DISCOLI), 2023.
- [W2] P. Hall[†], **J. Diller**, A. Moon[†], and Q. Han. “*DroNS-3: Framework for Realistic Drone and Networking Simulators*.” In proceedings of the Workshop on Micro Aerial Vehicle Networks, Systems, and Applications (DroNet), 2023.
- [W3] **J. Diller**, P. Hall[†], C. Schanker[†], K. Ung[†], P. Belous[†], P. Russell[†], and Q. Han. “*ICCSwarm: A framework for Integrated Communication and Control in UAV Swarms*.” In proceedings of the Workshop on Micro Aerial Vehicle Networks, Systems, and Applications (DroNet), 2022.

Extended Abstracts & Demonstrations

- [A1] M. Hatch[†], C. Felling[†], **J. Diller**, and Q. Han. “*ROARQuad: Robust, Open Academic Research Quadcopter*.” Accepted for demonstration at the International Conference on Mobile Systems, Applications, and Services (Mobisys), 2025.
- [A2] **J. Diller**. “*Self-Aware High-Level Planning for Robot Teams*.” Presented at the Cyber-Physical Systems Rising Stars Workshop, 2024. *Acceptance rate: 16.4%*.
- [A3] **J. Diller**. “*Planning and Coordination for Unmanned Aerial Vehicles*.” Presented at the International Conference on Autonomous Agents and Multiagent Systems’ Doctoral Consortium, 2023.
- [A4] **J. Diller**, Q. Han, C. Dreyer, F. Rossi, S. Bandopadhyay, J.P. de la Croix, A. Rahmani, P. Clark. “*Integrated Communication and Controls for Swarms of Small Satellites*.” Presented at the Inter-Planetary Small Satellite Conference, 2021.

† Student mentees

Awards & Recognitions

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| • 2024 Cyber-Physical Systems Rising Star | May, 2024 |
| • Best PhD Poster, C-MAPP, <i>Colorado School of Mines</i> | Jan, 2024 |
| • Best PhD Poster, C-MAPP, <i>Colorado School of Mines</i> | Jan, 2023 |
| • Best Elevator Pitch, C-MAPP, <i>Colorado School of Mines</i> | Feb, 2022 |
| • Graduation Student Marshal for School of Sci., Eng. and Tech. | May, 2020 |
| • Computer Science Outstanding Student, <i>Pennsylvania State University</i> | Apr, 2020 |
| • Evan Pugh Scholar Award - Senior, <i>Pennsylvania State University</i> | Apr, 2019 |
| • Evan Pugh Scholar Award - Junior, <i>Pennsylvania State University</i> | Apr, 2018 |

- President's Freshman Award, *Pennsylvania State University* Apr, 2017
- Dean's List, *Pennsylvania State University* Spring 2016 - Spring 2020

Teaching Experience

College Teaching Certificate Program August 2024 – May 2025

Colorado School of Mines, Trefny Center Golden, CO

- Studied teaching pedagogy and learner-centered teaching practices
- Designed an upper-level undergraduate course to introduce students to robot autonomy

Primary Instructor August 2024 – May 2025

Colorado School of Mines Golden, CO

- CSCI 128: Computer Science for STEM (Fall 2024, Spring 2025)
- Typical class size: 50-70 students

Teaching Assistant August 2020 – December 2022

Colorado School of Mines Golden, CO

- CSCI 565: Distributed Systems (Fall 2022)
- CSCI 406: Algorithms (Spr 2021)
- CSCI 261: Programming Concepts (Fall 2020)

Peer Tutor August 2017 – May 2020

Pennsylvania State University Middletown, PA

- Tutor students in Computer Science, Mathematics, Physics and Engineering classes.
- Earned CRLA Certified Tutor, Level I Certification.

Mentoring Experience

Graduate Students degree (year conferred)

Matthew Desaulniers, <i>publications</i> : [R1]	<i>PhD in Robotics</i>
Robert Byers, <i>publications</i> : [C1, C2]	<i>MS in Computer Science</i>
Yee Shen Teoh, <i>publications</i> : [C1]	<i>MS in Robotics (2025)</i>
Priestly Barigala	<i>MS in Robotics (2024)</i>

Undergraduate Students

Cody Felling*, <i>publications</i> : [A1]	<i>BS in Electrical Engineering (2025)</i>
Matthew Hatch*, <i>publications</i> : [A1]	<i>BS in Electrical Engineering (2025)</i>
Jack Sigler	<i>BS in Computer Science (2025)</i>
Ava Moon, <i>publications</i> : [W2]	<i>BS in Computer Science</i>
Corey Schanker, <i>publications</i> : [W3]	<i>Combined BS in CS + EE</i>
Peter Hall*, <i>publications</i> : [C4, W2, W3]	<i>BS in Computer Science (2023)</i>
Kristen Ung*, <i>publications</i> : [W3]	<i>BS in Electrical Engineering (2022)</i>
Philip Belous, <i>publications</i> : [W3]	<i>BS in Computer Science (2022)</i>
Zachary Smeton*	<i>BS in Computer Science (2021)</i>

* Graduated with Undergraduate Research Scholar Distinction

Work Experience

Robert Bosch GmbH

May 2018 – July 2020

Embedded Software Developer & Test Intern

Lancaster, PA

- Develop firmware for embedded real-time environments.
- New product prototyping.
- Design and implement automated tests for moving cameras.
- Write documentation for internal procedures.

United States Marine Corps

November 2010 - November 2015

KC-130J Crewmaster, Plane Captain

San Diego, CA & Okinawa, Japan

- Last held rank/pay grade: Sergeant/E-5
- Supervised small teams in pre and post flight inspections on C-130J aircraft.

Service

Technical Reviewer

- IEEE Robotics and Automation Letters (1 paper)
- Pervasive and Mobile Computing Journal (2 papers)
- ACM Journal on Autonomous Transportation Systems (1 paper)
- IEEE International Conference on Pervasive Computing and Communications (3 paper)
- IEEE International Conference on Robotics and Automation (1 paper)
- IEEE/RSJ International Conference on Intelligent Robots and Systems (1 paper)
- ACM/IEEE International Conference on Cyber-Physical Systems (1 paper)
- International Conference on Unmanned Aircraft Systems (1 paper)

Committees

- Graduate Research and Discovery Symposium, Program Chair 2025
- Graduate Research and Discovery Symposium, Poster Chair 2024
- Research Integrity and Security Committee, Grad. Rep. Fall 2023 - Spring 2024
- Research Council, Graduate Representative Spring 2023
- CS Research Committee, Student Advocate Spring 2022 - Spring 2023

Other Service

- Graduate Student Government, CS Representative Fall 2022 - Spring 2024
- Graduate Student Seminar Organizer:
 - 6 seminars in 2023
 - 4 seminars and 1 workshop in 2024
 - 2 seminars in 2025

References

Qi Han

Professor, Colorado School of Mines

qhan@mines.edu

Kevin Moore

Emeritus Professor, Colorado School of Mines

kmoore@mines.edu

John G. Rogers

Senior Research Scientist, DEVCOM Army Research Laboratory

john.g.rogers59.civ@army.mil